10.015 Physical World: Design 1D – Worksheet 1

Group :1

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1. **Describe and briefly explain your choice of mechanical energy sources to power the toy car and the design of the engine. Give an estimated amount of energy stored that is required of the toy car to complete the obstacle course. The obstacle course contains tracks with rough surfaces, ramps with specific height etc. Check out the details of the obstacle course in the 1D writeup.**

The mechanical energy that we chose is a rubber band powered mechanism. The rubber band acts as an energy storage when it is twisted or pulled. After the release, the energy is released and results in a force that will move the car forward as a reaction from the previous action.

This method was chosen after considering various other ways, namely
1. Springs: A pullback mechanism where the spring is attached to the back of the car and is stretched by pulling the car back.
2. Air – By using compressed air to expel air at the back of the car, pushing the car forward
3. Rubber Bands- Twisting the Rubber band by using a driver shaft attached to the rear wheels of the car. As the car is pulled back, the rubber band twists more, creating increasing elastic potential energy stored in the rubber band. The car moves forward when released.

The most effective way to choose the best method was by analyzing the advantages, disadvantages and the feasibility for each method.

Methods	Advantages	Disadvantages	Feasibility
Springs	- Larger the spring the better the efficiency - Durable as it can withstand multiple compression - lightweight	- May loose its elasticity the more the spring is stretched - Larger spring harder to obtain	- Modification that can be made with spring is limited. - Cost effective - Limited speed control
Compressed Air	- Less wear and tear on components. - High speed potential. Compressed air generates a powerful burst of	- Difficulty in maintaining a straight direction - Potential for sudden pressure drop	- Hard to modify after creation - Expensive to remodel if prototype fails - Multiple cartridges needed when

	energy, allowing the car to reach high speeds.		testing out our model which can become expensive and exceed our budget. - Potential for air leakage - The need for an air tank limits car space to place the weights in the car. - Multiple resources are required to make the car air tight.
Rubber Bands	- Easy to use - Both the thin anted thick rubber bands are easy and cheap resources to obtain. - Lightweight - Inexpensive - Reusable	- Wear and tear of rubber bands are more common - Controlling speed and direction of car can be difficult	- Set up for a rubber band car is easier to modify (add/remove rubber bands or increase/decrease thickness of rubber bands) - Any modifications made to the model is cheaper

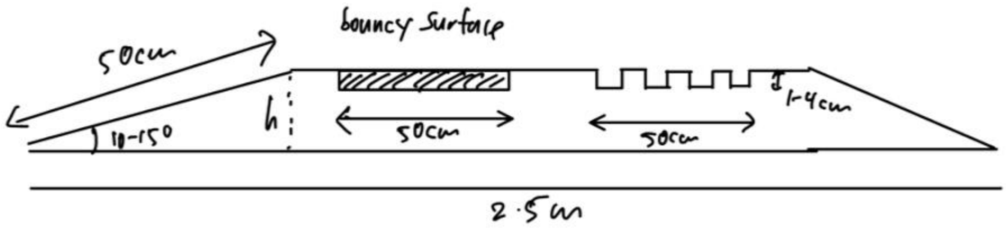
Through this analysis, we found that using the rubber band technique is most suitable as resources are most accessible and cheaper compared to other techniques. Although there are challenges in controlling direction and speed of the car, this issue can be tackled by designing our car such that it can overcome these challenges

To calculate the energy, the potential energy formula can be used, where k is the spring constant of the rubber band, and x is the extension of the rubber band.

$$EP = \frac{1}{2} k\, x^2$$

K=35.5N/m
https://www.ijres.org/papers/Volume-10/Issue-3/Ser-1/F10032838.pdf

The spring constant K of the rubber band used can be determined experimentally, by plotting a graph of Force against extension x. Based on Hooke's law, within the limit of proportionality, the gradient of the graph represents the spring constant k.



By conservation of energy

$$EPE = KE + GPE + W_{friction}$$

$$\frac{1}{2}kx^2 = \frac{1}{2}mv^2 + mgh + f \cdot s$$

$$= \frac{1}{2}mv^2 + mg(0.5\sin 15°) + \mu_k N \cdot 0.5 + \mu_s N \cdot 0.5$$

μ_k = coefficient of rough surfaces

μ_s = coefficient of smooth surfaces

∴ for the car to have enough energy, both the spring constant k and extension x must be maximised.

* main challenges to overcome:

- getting up the ramp
- overcoming rough surfaces
- staying on track

Assumptions: $k = 35.5\ Nm^{-1}$, $x = 0.3\ m$, $m = 1\ kg$, $g = 9.81$
$\mu_s = 0.6$, $\mu_k = 0.9$, $v = 0.5\ ms^{-1}$

GPE up slope $= mgh = (1)(9.81)(0.5 \sin 15)$
$= 1.270\ J$

kinetic friction required to overcome $= \mu_k N + \mu_s N$
$= \mu_k mg + \mu_s mg$

total work done against friction $w = f \cdot s$
$= 0.5(0.6 \cdot 1 \cdot 9.81 + 0.9 \cdot 1 \cdot 9.81) = 7.358\ J$

kinetic energy to sustain $= \frac{1}{2}mv^2$
$= \frac{1}{2} \cdot 1 \cdot 0.5^2$
$= 0.125\ J$

$\therefore$ total energy required $= 1.270 + 7.358 + 0.125$
$= 8.753\ J$

$EPE = 8.753$
$\frac{1}{2}kx^2 = 8.753$
$x = \sqrt{\frac{2 \cdot 8.753}{35.5}} = 0.7\ m$

extension required $\approx 0.7\ m$

2. **Briefly explain your plan to deliver the energy to the wheels to move the toy car.**

A rubber band is attached to the driver shaft. The mechanism works by twisting the back wheels. The elastic potential energy that is stored in the rubber band will be converted into kinetic energy in the wheels at the drive shaft once the rubber band is released. The kinetic energy of the wheels of the drive shaft will cause the toy car to move, providing an initial driving force for the toy car. The front wheels of the car will also start to move and are used to control the general direction of the acceleration of the toy car.

$$EP = KE + GPE$$
$$\frac{1}{2}kx^2 = \frac{1}{2}mv^2 + mgh$$

Where m is the mass of the car, v is the velocity gained by the car, g is gravitational constant, and h is the height of the ramp.
The spinning of the axle caused by the rubber band generates torque in the wheels. The torque generated causes the wheel to spin, allowing the car to move.

3. **Explain and describe the engineering solutions you consider ensuring the toy car is able to overcome the challenges such as the upward and downward vertical step (1 cm – 4 cm).**

- **Bigger wheels**
  Larger wheels generate more torque than smaller wheels (given by the equation, $\tau = F\,x\,D$, where τ is the torque, F is the force and D is the diameter), which enables it to generate enough force to overcome obstacles such as the small bumps and gaps in surfaces. Larger wheels also increase the clearance of the car, increasing the approach angle of the car, allowing it to overcome obstacles.

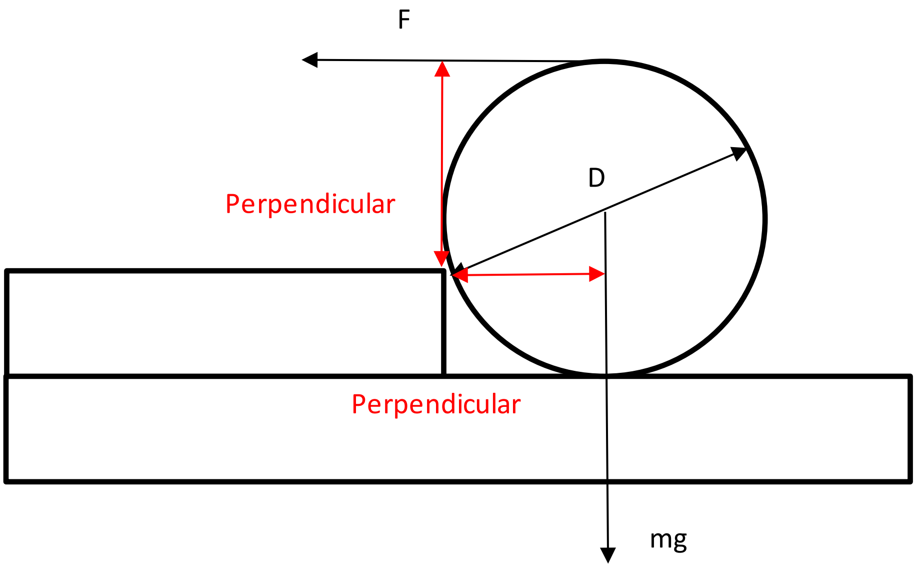


Fig 1.1: Torque required to overcome gaps

- low center gravity.A lower center of gravity ensures that the car will be less likely to topple over and stay on track of the obstacle.
- **Reducing friction in the wheels:**
  - Use of ball bearing at the front axle
    Ball bearing reduces friction between the axle and the wheels such that the wheels spin more smoothly. It also enables the wheels to spin faster, which can lead to higher speeds.
  - Use of washers between axle
    wheel and the car frame – prevents the wheels from touching the car frame. The washers also ensure that the wheels can continuously spin the same direction, preventing the car from veering off.
  - Use of grease/ WD40 (KIV)
    Using grease is optional to ensure smoother motion for ball bearings and axle

4. **Explain your plan to ensure there is sufficient tractions and stability as the car moves (moving in straight line without toppling and wheel slipping easily, able to face bouncy surfaces) on the obstacle course.**

- **High traction tires (slick tires)**
  The kinetic friction between the base of the wheels and the ground surface is given by μN where μ is the friction coefficient and N is the normal contact force between the wheel and the ground. Increasing the traction, specifically using slick tires increases this friction

coefficient, thus increasing the kinetic friction between the tires and the surface, preventing the car from veering off course.

- **Wider wheels (thickness of the wheel)**
  Wider wheels increase the surface area of contact with the ground. This prevents the center of gravity from falling outside the base of the car, keeping the car within the track.
- **Washers**
  Washers are like fasteners that are placed between the axle and wheels. The function of it is mainly for securing and ensuring the wheels do not spin freely. Washers are also used for limiting the sideway movement, preventing the wheels from shifting side to side.
- **Open wheel car**
  The wheel placement outside the car's main body results in faster acceleration and higher speed. The wheels are not moving in a limited are of which the body of the car is encasing, which enables the car to move easily compared with a closed wheel car.

5. **Provide simple, clear and informative schematic diagrams (top view and side view) of your design, label all key components, dimensions, and materials of choice. You may consider using CAD (not compulsory) if you want.**

Dimensions :

- Wheel – inner radius 11mm, total radius 40mm, 20mm width
- Ball bearing inner radius 5mm, outer radius 11mm
- Car width 70mm, 175mm length
- Axle length 12cm, axle radius 5mm

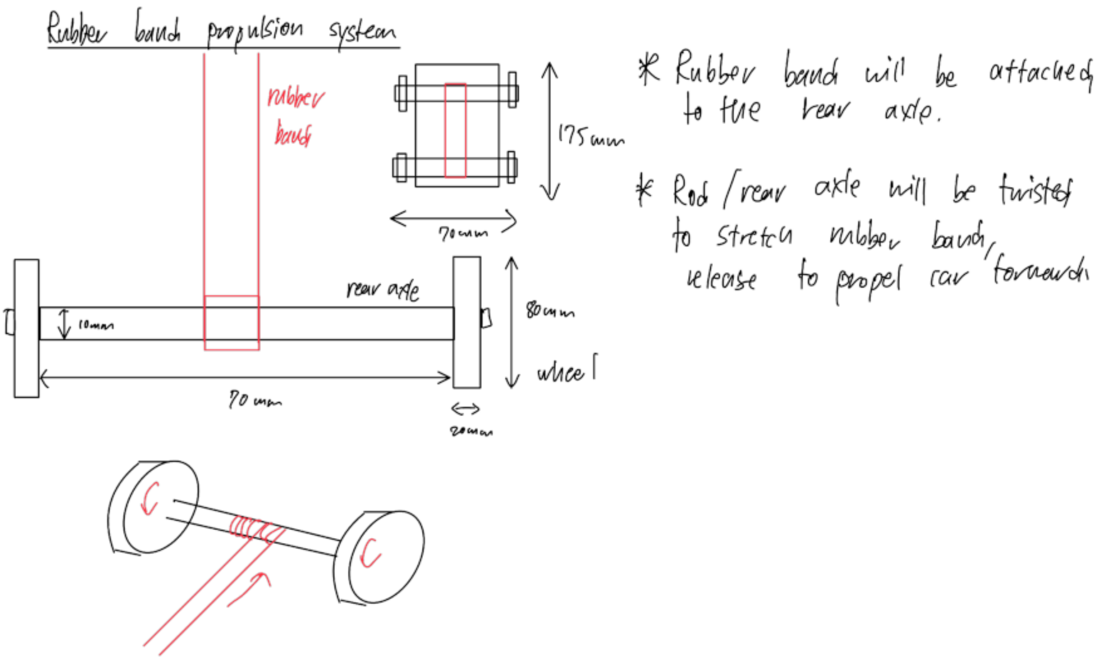


Fig 2.1: Rubber band placement in car sketch

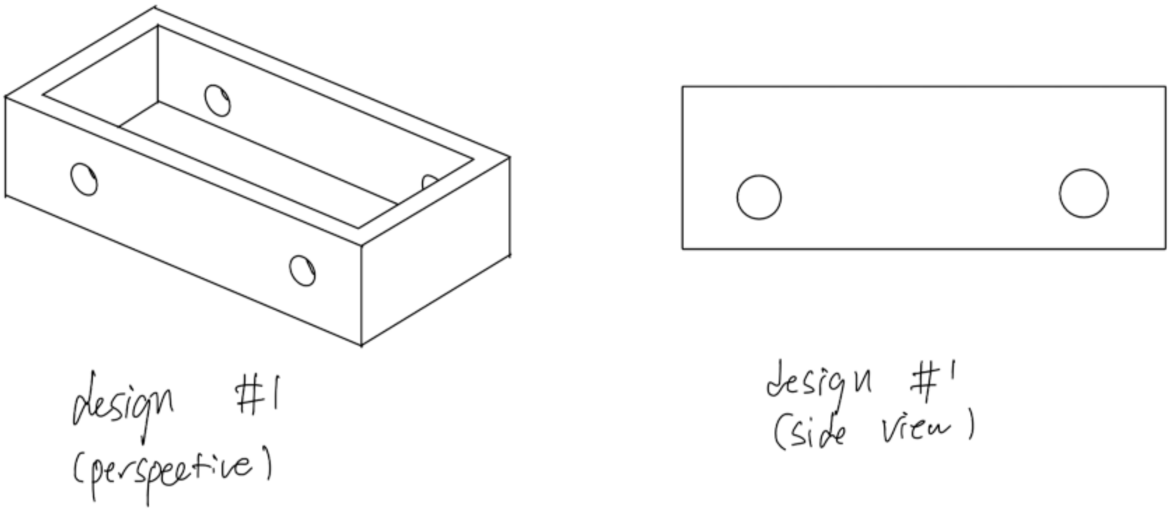


Fig 2.2: Design 1 sketch

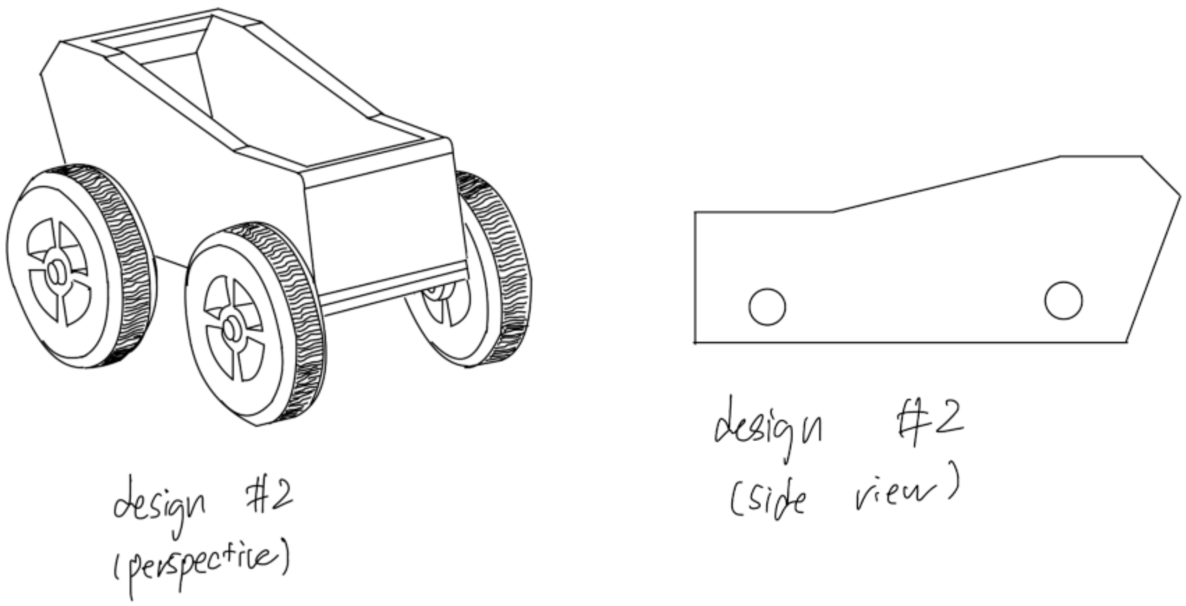


Fig 2.3: Design 2 sketch (Including wheels)

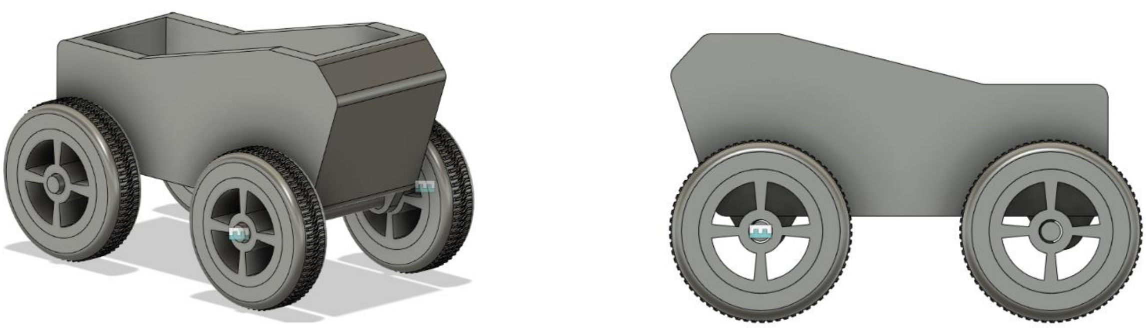


Fig 2.4 CAD Design 2 in fusion 360